

DSPIRA 2018 LNA assembly

Pranav Sanghavi | June 22, 2018

Institute reference. Match these component values to your board revision.

Refer to the schematic and layout in Figures 1 and 2 while assembling the LNA.

Assembly

Assemble all surface mount passives first: resistors, capacitors, and inductors. Work from left to right on the board. Install components with the same value together.

Component Label	Value
L2	10 nH
L1, L3, L4, L5, L6	100 nH
C6	9.1 pF
C1, C5	27 pF
C15	6.8 pF
C4	4700 pF
C2, C3, C7, C8, C9, C10, C11, C12, C13, C14	330 pF
R5	4.7k Ω
R2	27k Ω
R1	33k Ω
R3	50 Ω
R8	150 Ω
R4, R13	43 Ω
R6	17.4 Ω
R7, R14	5.8 Ω
R9, R10	287 Ω
R11, R12, R15, R16	870 Ω

Now place the actives:

Component label	Value	Notes
U2	SAV-541	Make sure the large pad lines up with the big foot
U1, U3	GALI-39+	
U4, U5	BFCN-1445	Align dot to point towards input
IC1	LM2940	

Figure 1: LNA circuit schematic

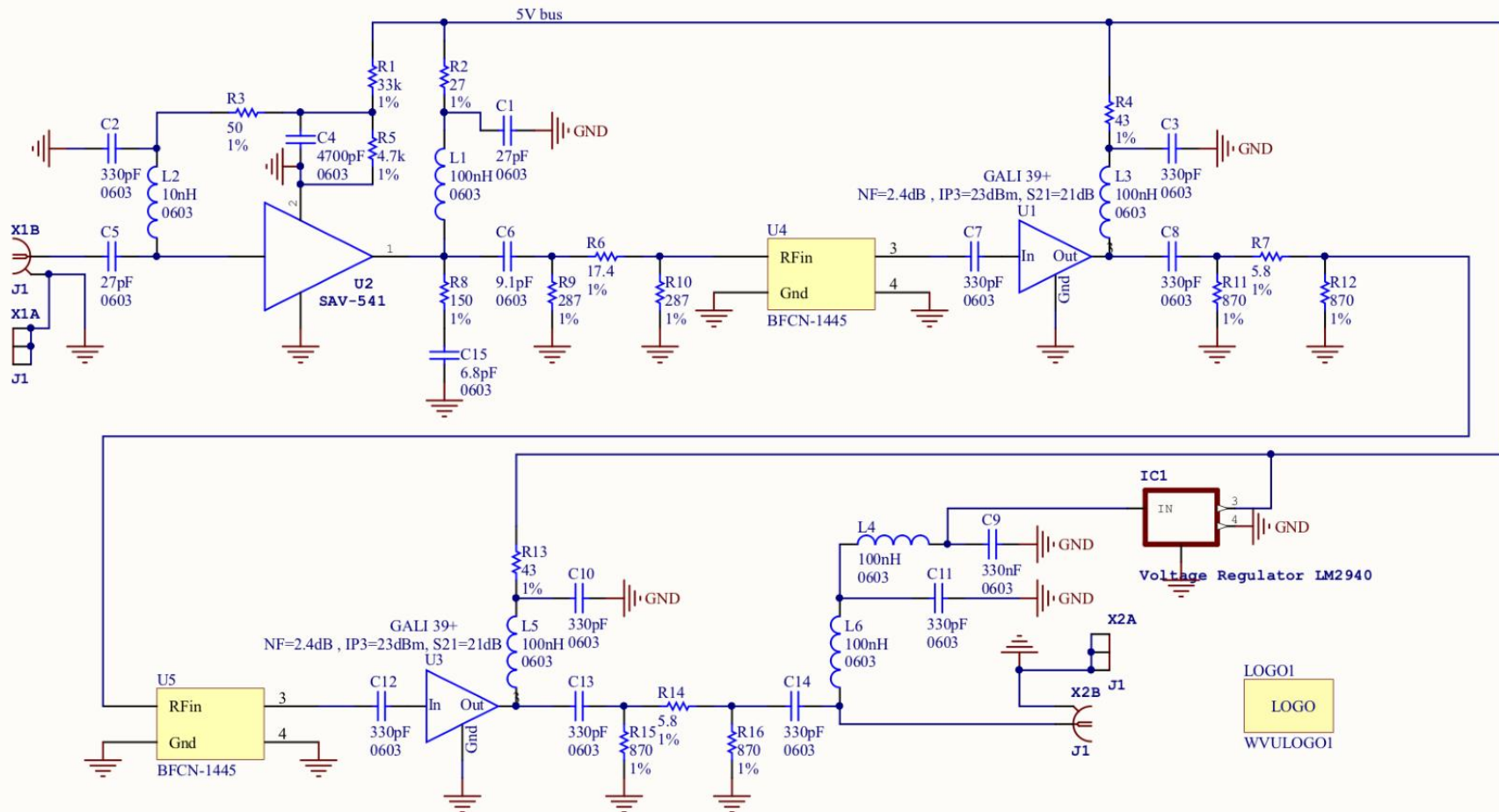


Figure 2: LNA board layout

